

CRM Code Map

A navigation guide for the **godrej-crm-streamlit** repository — find the right file to change when you want to fix or extend any feature.

How to use this guide

Every file in the cleaned repo is listed below, grouped by purpose. Each entry tells you (a) what the file does, (b) when to open it, and (c) what other files it depends on so you know what else to check after a change. The repo has two independent runtimes — they share the `services/` layer but run separately:

- 1. The Streamlit dashboard** — entry point `streamlit_app/app.py`. Everything in `pages/` is a screen in the sidebar.
- 2. The scheduled jobs** — the `*_job.py` files at the `streamlit_app/` root. Each is invoked by a YAML workflow in `.github/workflows/` on a cron schedule (no Streamlit UI; pure Python).

Mental model — request flow

User opens a page	→	<code>pages/<file>.py</code>	→	<code>services/<helpers></code>	→	Google Sheet / Gmail
GitHub Actions cron	→	<code><name>_email_job.py</code>	→	<code>services/email_sender_*</code>	→	Gmail SMTP
GitHub Actions cron	→	<code><name>_import_job.py</code>	→	<code>services/imap_lead_import / mis_email_import</code>	→	Google Sheet

1. Entry points

Files that boot a runtime. Start here if you're confused about where execution begins.

streamlit_app/app.py

What it does: Navigation controller for the entire Streamlit app. Defines the sidebar page list via `st.navigation()`, sets the page title / icon, and exposes the *Show Old Data Dashboard (Pre FY 26-27)* sidebar toggle.

Open this file when you want to: add or remove a sidebar entry, change the default landing page, rename a screen, change the app title / icon, or toggle the historical dashboard.

Talks to: Imports every file in `pages/`.

streamlit_app/scheduler.py

What it does: Local-dev fallback for the cron jobs. Uses the `schedule` library to fire pending-delivery and MIS-import jobs in process. Production runs these via GitHub Actions instead.

Open this file when you want to: test the email/import flow on your own laptop without waiting for GitHub Actions, or debug a job interactively.

Talks to: `services.email_sender`, `services.sheets`.

2. Dashboard pages — what you see in the sidebar

One file per screen. The number in the filename controls sort order but isn't shown to the user — the title in `app.py` is.

`streamlit_app/pages/b2c_dashboard.py`

What it does: ■ **4sInteriors B2C Sales Dashboard** — the landing page. Combines Franchise + 4S sheets (FY 26-27 column format) and shows live headline KPIs, pending-delivery alerts, WhatsApp share buttons, and the *Send Email* buttons for pending-delivery / update-delivery emails.

Open this file when you want to: fix anything on the home screen — KPI tiles, pending-delivery section, WhatsApp links, or the manual *Send Email* buttons on this page.

Talks to: `services.sheets`, `services.automation4s`, `services.email_sender_4s`, `services.delivery_updates`, `services.dashboard_ticker`.

`streamlit_app/pages/daily_b2c_sales.py`

What it does: ■ **Daily B2C Sales** — per-day, per-salesperson revenue tracker with the FY 26-27 Sales Target & Achievement view (the screen that replaced the old `80_Sales_Targets.py`). Uses *CROSS CHECK GROSS AMT* as the revenue column.

Open this file when you want to: change daily revenue calculations, per-salesperson breakdowns, monthly target editing, or Plotly target-vs-actual charts.

Talks to: `services.sheets`, `services.incentive_store`.

`streamlit_app/pages/17_Customer_Intelligence_Engine.py`

What it does: ■ **Customer Intelligence Engine** — cohort analysis (High-Value, Loyal, At-Risk, Dormant, New, Full List) that spans BOTH current and historical sheets. Each customer row exposes WhatsApp *App* and *Web* deep-links.

Open this file when you want to: tweak cohort thresholds (■5L for High-Value, 90-180d for At-Risk, etc.), the per-cohort columns shown, or the WhatsApp message template.

Talks to: `services.sheets`.

`streamlit_app/pages/20_Product_Sales_Analysis.py`

What it does: ■ **Product Sales Analysis** — category (Home Storage / Home Furniture / Others) and sub-category breakdown. Handles the special *Kreation X3* modular-unit math (count Main + Dresser rules) so unit-counts are correct.

Open this file when you want to: change product categorisation, the *Kreation X3* unit-counting rule, or add a new sub-category auto-detection keyword.

Talks to: `services.sheets`.

`streamlit_app/pages/30_Sales_Reports_and_Strategy.py`

What it does: ■ **Sales Reports and Strategy** — BA-grade analytical dashboard: headline KPIs, monthly target tracker, day-of-week / hour-of-week heat-maps, salesperson leaderboard + GMB ratings, category mix, customer cohort summary, and the *Odia festival calendar* with strategy recommendations.

Open this file when you want to: edit any analytical chart, add a new KPI tile, change leaderboard rules, or update the festival-calendar source URL / fallback list.

Talks to: `services.sheets`.

`streamlit_app/pages/40_Products_catalog.py`

What it does: ■ **Products Catalog** — paginated catalog viewer for the *Product Catalog* and *Discontinued Products* sheets. 10 items per page.

Open this file when you want to: change catalog filters, pagination size, the discontinued-products view, or the catalog image-rendering logic.

Talks to: Reads Google Sheets directly (does not go through `services.sheets`).

streamlit_app/pages/50_MIS_Update.py

What it does: ■ **MIS Update** — viewer for the cached *BR_MIS* data (the daily Godrej MIS Excel attachment fetched from Gmail at 11 AM). Rows where *Sales Order Qty* == *Sales Order Committed Qty* are highlighted green = ready for delivery.

Open this file when you want to: change MIS columns displayed, the green-highlight rule, or the manual *Refresh* button behaviour.

Talks to: `services.sheets`, `services.mis_email_import`.

streamlit_app/pages/70_Leads.py

What it does: ■ **Leads** — lead pipeline screen. Lists leads from JustDial / IndiaMART / Facebook / web that the IMAP job auto-imports, lets the team add follow-ups, change status, and convert leads.

Open this file when you want to: change lead-status options, follow-up date logic, the lead-add form, or the lead-detail layout.

Talks to: `services.sheets`.

streamlit_app/pages/90_Sales_Team_Tasks.py

What it does: ■ **Sales Team Tasks** — the screen that renders the *SALES_TEAM_TASK* master sheet as per-day occurrences. Supports Daily / Adhoc / Weekly / Monthly frequencies and respects employee week-off maps. Has add / edit / mark-done controls inline.

Open this file when you want to: change task expansion rules, add a new frequency, modify the inline edit form, or fix how week-offs interact with task generation.

Talks to: `services.sheets`, `services.sales_task_expander`.

streamlit_app/pages/95_Happy_Calling.py

What it does: ■ **Happy Calling** — list of delivered customers with no Happy Calling Date logged. Lets sales staff log the call inline. Mirrors the data the 7 AM Happy Calling email shows.

Open this file when you want to: edit the default date filter (1 Apr 2026 → today), add a remarks column, or change the inline date-logging widget.

Talks to: `services.happy_calling`.

streamlit_app/pages/100_Sales_Manager_Dashboard.py

What it does: ■ **Sales Manager Dashboard** — the full Sales Incentive engine for Q1 FY 26-27. Tangible block (70%): quarterly tier 0.5/1/1.25%, Rs 10K consecutive-month bonus, 0.5% B2B incentive, locker incentive. Star block (30%): HF-HS, B2B-stars, upsell, leads ×3, NPI ×2, repeat customers, 5-star reviews, COO admin, grooming, attendance < 98%, in/outdoor activity. ADMIN / MANAGER / OWNER / PROPRIETOR only, with a dedicated *Incentive_Users* bcrypt sheet.

Open this file when you want to: change any incentive formula, tier breakpoint, star rate (■20/star), seed new quarterly targets, or audit who's hit what.

Talks to: `services.sheets`, `services.incentive_store`.

streamlit_app/pages/old_data_dashboard.py

What it does: ■ **Old Data Dashboard (Pre FY 26-27)** — hidden by default, shown only when the sidebar toggle is on. Same shape as the live dashboard but reads from *OLD_SHEET_DETAILS* (legacy column format) and uses the `legacy_automation.py` / `email_sender.py`.

Open this file when you want to: look at historical pre-FY-26 data or fix something that only the old format triggers.

Talks to: `services.sheets`, `services.automation`, `services.email_sender`.

3. Services — shared business logic

Everything in `streamlit_app/services/`. Both the dashboard pages and the scheduled jobs import from here, so a bug fix here usually fixes the issue in multiple places at once.

3a. Data access

`streamlit_app/services/sheets.py`

What it does: The **Google Sheets data layer**. Owns the spreadsheet ID, the gspread client, the `get_df()` / `write_df()` / `_get_spreadsheet()` primitives, plus helpers like `upsert_record`, `update_followup`, `append_email_log`, and `was_email_sent_today`. Imports user-management helpers from `sheets_1.py` for backward compatibility.

Open this file when you want to: change which Google sheet is read, the auth flow, the caching TTL, or any low-level read/write behaviour. **Every dashboard page and every email job depends on this — touch carefully.**

Talks to: `utils.helpers`, `services.sheets_1`.

`streamlit_app/services/sheets_1.py`

What it does: Holds the user-management helpers (`get_users_df`, `upsert_user`, `ensure_users_header`, `deactivate_user`) and is re-exported by `sheets.py`. Kept separate so old import paths still work.

Open this file when you want to: add / edit / deactivate CRM user accounts at the data layer, or change the Users sheet schema.

Talks to: Independent — only imported by `services.sheets`.

`streamlit_app/services/auth.py`

What it does: Login / role authorisation. Roles are Viewer < Editor < Admin. Password hashing uses `bcrypt`. Exposes `AuthService` and `current_user_badge`.

Open this file when you want to: change role policies, password rules, or the login-form rendering.

Talks to: `services.sheets`.

`streamlit_app/services/incentive_store.py`

What it does: Reads / writes the three incentive sheets — *Incentive_Quarterly_Targets*, *Incentive_Audit_Log*, *Incentive_Users*. Seeds Q1 FY 26-27 targets if empty. 60-second read cache. Every page load + filter change is audit-logged.

Open this file when you want to: change incentive target seeds, audit-log columns, the cache TTL, or the way *Incentive_Users* credentials are stored.

Talks to: `services.sheets`.

3b. Delivery + MIS reconciliation

`streamlit_app/services/delivery_readiness.py`

What it does: Cross-references the cached MIS sheet with *All Sales Records* to decide which Pending / Overdue rows are **READY** for delivery (SO Qty == SO Committed Qty for every line item). Exposes `is_item_ready`, `ready_so_set`, `ready_godrej_so_for_orders`, `ready_so_lookup`.

Open this file when you want to: change the green-highlight rule on the dashboard or in the delivery emails.

Talks to: Pure logic — depends on dataframes only.

`streamlit_app/services/delivery_updates.py`

What it does: Logs every dashboard-side delivery-date / remarks update into the *Pending Delivery Updates* Google sheet. Captures who updated, when, and whether the customer was informed. This is the audit trail of all pending-delivery edits.

Open this file when you want to: change which columns are logged, who's allowed to update, or the audit-row schema.

Talks to: `services.sheets`.

streamlit_app/services/mis_email_import.py

What it does: Logs into Gmail via IMAP, finds today's *BR_MIS - Interio MIS (4S INTERIO)* email, parses the *PO* sheet of the Excel attachment, and caches the cleaned DataFrame into the *MIS_Daily* Google Sheet. Exposes `fetch_and_cache_mis`, `load_cached_mis`, and the cache-sheet name.

Open this file when you want to: fix MIS column mapping, change the email subject we search for, edit the row filtering, or change the cache TTL.

Talks to: `services.sheets`.

3c. Leads + tasks + happy-calling

streamlit_app/services/imap_lead_import.py

What it does: Reads new leads from *4sinteriorsbbsr@gmail.com* via IMAP and writes them into the LEADS sheet. Recognises JustDial / IndiaMART / Facebook / Sulekha / web-form / direct-email formats and de-duplicates against existing leads.

Open this file when you want to: add a new lead-email parser, change duplicate-detection rules, or fix the phone-number normalisation.

Talks to: `services.sheets`.

streamlit_app/services/sales_task_expander.py

What it does: Expands the SALES_TEAM_TASK master sheet (templates only) into per-day, per-employee task rows the same way the dashboard does — for both the dashboard and the email jobs. Honours week-offs. Exposes `expand_with_status`, `get_missed_tasks`, `get_today_and_overdue`, `load_employee_weekoff_map`.

Open this file when you want to: change how recurring tasks are generated, add a new FREQUENCY type, or fix the assignment of multi-person tasks (comma/&-separated names).

Talks to: Pure logic over dataframes.

streamlit_app/services/happy_calling.py

What it does: Domain logic for Happy Calling — what counts as *pending happy call*, what the unique key is (ORDER NO, falling back to CUSTOMER NAME + DELIVERY DATE), and the join with the *Happy Calling Sheet*. Shared by the dashboard page and the 7 AM email job.

Open this file when you want to: change happy-call eligibility, the keying rule, or the duplicate-detection.

Talks to: `services.sheets`.

3d. WhatsApp + alerts + dashboard ticker

streamlit_app/services/automation4s.py

What it does: Used by the live FY 26-27 *b2c_dashboard*. Builds alert payloads (pending delivery, overdue, missing dates), generates the WhatsApp deep links (what sapp : // send for the desktop/mobile app, web.what sapp . com / send for the browser), and formats tabular WhatsApp messages.

Open this file when you want to: change WhatsApp message templates, alert thresholds, or the column names the alert engine looks at (the FY 26-27 names).

Talks to: Pure logic over dataframes.

streamlit_app/services/automation.py

What it does: Legacy version of the alert + WhatsApp logic — only used by `old_data_dashboard.py` (pre FY 26-27 column names).

Open this file when you want to: fix something that only happens on the old historical dashboard.

Talks to: Pure logic over dataframes.

streamlit_app/services/dashboard_ticker.py

What it does: Builds and renders the right-to-left marquee at the top of the dashboard. Surfaces things that need attention *today* — pending deliveries, overdue rows, missing delivery dates, payment due, happy-calls done, sales-team task health, lead pipeline. Pure HTML/CSS with `@keyframes` animation.

Open this file when you want to: change what's shown in the ticker, the colour, the speed, or pause-on-hover.

Talks to: `services.sheets`.

3e. Email senders

streamlit_app/services/email_sender_4s.py

What it does: The main email sender used by the live FY 26-27 setup. SMTPs out five emails: Pending Delivery (10 AM), Update Delivery Status (11 AM), Pending Delivery evening repeat (5 PM), Payment Due Morning (10 AM), Payment Due Reminder (11 AM). Credentials cascade env vars → Streamlit secrets → `.env`.

Open this file when you want to: change email subject lines, HTML templates, the recipient list, or the rules for which rows go into each report.

Talks to: `services.sheets`.

streamlit_app/services/email_sender.py

What it does: Legacy email sender used only by the in-process scheduler `.py` and `old_data_dashboard.py`.

Open this file when you want to: fix the legacy (pre FY 26-27) email paths.

Talks to: —

streamlit_app/services/email_sender_delivery_schedule.py

What it does: Composes the *Schedule Delivery* email triggered when an operator ticks ready-for-delivery rows. Pulls every line item from the cached MIS for the matched GODREJ SO NO, formats them as a numbered list, and attaches the customer invoice PDF from Google Drive (`<Drive folder>/<Month Year>/<Day>/<Customer>.pdf`).

Open this file when you want to: change the schedule-delivery email body, the Drive folder layout, the invoice-matching rule, or the *Same-day Delivery and installation* tag.

Talks to: `services.sheets`, `services.delivery_readiness`.

streamlit_app/services/email_sender_happy_calling.py

What it does: Composes and sends the daily 7 AM Happy Calling email — customers whose delivery is done but no Happy Calling Date logged.

Open this file when you want to: change the 7 AM email's body, recipient list, or the cutoff date.

Talks to: `services.happy_calling`.

streamlit_app/services/email_sender_sales_tasks.py

What it does: Two daily emails: 11 AM (today's + overdue tasks, grouped by Daily / Adhoc / Weekly / Monthly) and 8 PM (status summary + per-task detail).

Open this file when you want to: change task-email templates, grouping, recipient list, or which task fields appear in the email.

Talks to: `services.sales_task_expander`.

streamlit_app/services/email_sender_monthly_performance.py

What it does: End-of-month (last day, 7 PM IST) per-salesperson performance email. Breaks each person's monthly tasks into Done / Pending / Overdue / Missed with the actual task names listed under each bucket.

Open this file when you want to: change the monthly-summary layout, the per-bucket task name list, or the way attendance / completion is rolled up.

Talks to: `services.sales_task_expander`, `services.sheets`.

3f. Google Business Profile (GMB) reviews

streamlit_app/services/google_reviews_service.py

What it does: Pulls reviews from the GMB v4 Reviews API (OAuth refresh-token auth), matches each reviewer to a 4S Sales row via a 5-priority matcher (email → phone → exact name → fuzzy name → token-overlap), and writes the star rating into the 4S Sales sheet's *REVIEW* column in a single batch update. Every run is audited to *REVIEW_SYNC_LOG* + *REVIEW_DETAILS*.

Open this file when you want to: change the matcher priority order, the fuzzy threshold, the target column, or any GMB auth detail.

Talks to: `services.sheets`.

4. Scheduled jobs (GitHub Actions)

Every job below is a standalone Python script invoked by a YAML workflow in `.github/workflows/`. They don't use Streamlit — they import from `services/` and SMTP / write to Sheets directly. If an email or import is firing at the wrong time, change the *cron* line in the workflow file, not the Python.

`streamlit_app/email_job.py`

What it does: Combined Delivery Alert Job. Runs twice daily — 07:00 IST (morning slot, subject *[4s CRM] Pending Delivery Alerts*) and 16:00 IST (evening slot, subject *[4s CRM] Update on Pending Delivery Alerts*). SLOT is set by the workflow; no hour-based fallback. Skips sending when both upcoming and overdue are empty.

Open this file when you want to: change pending-delivery email timing, subject lines, or the empty-skip rule.

Talks to: `services.sheets`, `services.email_sender_4s`. Workflow: `.github/workflows/godrej-email.yaml`.

`streamlit_app/payment_email_job.py`

What it does: Payment Due Email Job. 08:00 IST [*SLOT=morning*] — Payment Due Morning ($PENDING_DUE > 0 + \text{delivery} \leq \text{yesterday}$). 09:00 IST [*SLOT=reminder*] — Payment Due Reminder ($PENDING_DUE > 0 + \text{delivery} \leq \text{today}$). Strict SLOT/MANUAL_JOB isolation — a misfired cron cannot send the wrong report.

Open this file when you want to: change payment-due thresholds, the delivery-date cutoffs, or the slot-isolation rules.

Talks to: `services.email_sender_4s`. Workflow: `.github/workflows/payment-email.yaml`.

`streamlit_app/happy_calling_email_job.py`

What it does: Daily 07:00 IST — Happy Calling email. List of customers whose delivery is DELIVERED but no Happy Calling Date logged yet. List keeps appearing every day until the call is logged on the CRM page.

Open this file when you want to: change Happy Calling email timing or its recipients.

Talks to: `services.happy_calling`, `services.email_sender_happy_calling`. Workflow: `.github/workflows/happy-calling-email.yaml`.

`streamlit_app/sales_tasks_email_job.py`

What it does: Daily 11:00 IST — Sales Team Tasks email. Today's + overdue tasks grouped by FREQUENCY (Daily / Adhoc / Weekly / Monthly). Uses `append_email_log + was_email_sent_today` to make duplicate sends impossible on the same day.

Open this file when you want to: change 11 AM task-email rules, grouping, or duplicate-suppression.

Talks to: `services.sales_task_expander`, `services.email_sender_sales_tasks`. Workflow: `.github/workflows/sales-tasks-email.yaml`.

`streamlit_app/sales_tasks_status_email_job.py`

What it does: Daily 20:00 IST — Sales Team Task Status email. Summary counts (total / done / pending / overdue) + per-task detail tables.

Open this file when you want to: change the 8 PM status email format or scope.

Talks to: `services.sales_task_expander`, `services.email_sender_sales_tasks`. Workflow: `.github/workflows/sales-task-status-email.yaml`.

`streamlit_app/monthly_performance_email_job.py`

What it does: Runs on the last 4 days of every month, but only actually sends mail when today is the last day (clean workaround for cron's lack of *last day of month*). Sends per-salesperson Done / Pending /

Overdue / Missed breakdowns with task names.

Open this file when you want to: change the last-day detection, the monthly-email cohorts, or the day-window the workflow fires within.

Talks to: `services.email_sender_monthly_performance`, `services.sales_task_expander`. Workflow: `.github/workflows/monthly-performance-email.yaml`.

streamlit_app/lead_email_import_job.py

What it does: Every 30 minutes (cron `*/30 * * * *`) — reads new lead-source emails from `4sinteriorsbbsr@gmail.com`, parses them, and appends to the LEADS sheet. This is what populates the Leads page automatically.

Open this file when you want to: change the import cadence, add a new source parser, or fix duplicate detection.

Talks to: `services.imap_lead_import`. Workflow: `.github/workflows/lead-email-import.yaml`.

streamlit_app/mis_daily_import_job.py

What it does: Once-a-day at 11 AM IST — fetches the BR_MIS Excel attachment from Gmail, parses the PO sheet, writes to the *MIS_Daily* Google sheet. Idempotent — safe to re-run any time.

Open this file when you want to: change MIS import timing, the email-subject search string, or the row filtering before write.

Talks to: `services.mis_email_import`. Workflow: `.github/workflows/mis-daily-import.yaml`.

streamlit_app/google_reviews_update_job.py

What it does: Nightly at 22:00 IST — fetches GMB reviews and writes the star ratings back into the 4S Sales sheet. Two auth sets: service-account JSON for Sheets, plus GMB_CLIENT_ID/SECRET/REFRESH_TOKEN + GMB_ACCOUNT_ID/LOCATION_ID for the v4 Reviews API.

Open this file when you want to: change review-fetch timing, switch GMB location, or fix the column write.

Talks to: `services.google_reviews_service`, `services.sheets`. Workflow: `.github/workflows/google-reviews-fetch.yaml`.

5. Utilities, config, and repo metadata

`streamlit_app/utils/helpers.py`

What it does: CRM-wide formatting helpers: `fmt_number` (integers → no decimals, floats → 2 dp trimmed), `fmt_amount` (■-formatted), and `standardize_columns` (used by every data load to upper-case column names + strip spaces).

Open this file when you want to: change ANY number-display rule across the CRM, or the column-normalisation rules used at data-load time.

Talks to: Imported by `services.sheets` and most pages.

`streamlit_app/config/credentials.json`

What it does: Google service-account JSON. Used only as a local-dev fallback when Streamlit secrets aren't available.

Open this file when you want to: rotate the local service account or fix permissions on the local dev box.

Talks to: —

`.streamlit/secrets.toml`

What it does: Live Streamlit secrets (Google service account, Gmail SMTP credentials, GMB OAuth, etc.). Never committed.

Open this file when you want to: rotate creds, update SMTP password, or change the recipient list used by all emails.

Talks to: Read by every `services/` file that needs auth.

`.env`

What it does: Local environment fallback for credentials (lowest priority). Used when neither env vars nor Streamlit secrets are set.

Open this file when you want to: set up local-dev credentials without touching Streamlit secrets.

Talks to: —

`requirements.txt`

What it does: Python dependency manifest used both by Streamlit Cloud and by GitHub Actions (`pip install -r requirements.txt`).

Open this file when you want to: add a new Python library, pin a version, or upgrade dependencies.

Talks to: —

`.github/workflows/*.yaml`

What it does: Nine GitHub Actions workflows — one per scheduled job above. Each *cron* line is in UTC; convert to IST when reading.

Open this file when you want to: change a schedule, the timeout, secrets passed in, or temporarily disable a job (set on: `workflow_dispatch` only).

Talks to: Each workflow runs a single `*_job.py`.

6. Feature → file index (reverse lookup)

If you know the feature, this table tells you which files to open.

CRM feature	Files to open
Home / B2C dashboard layout & KPIs	pages/b2c_dashboard.py
Daily sales / target tracker	pages/daily_b2c_sales.py + services/incentive_store.py
Customer cohorts (High-Value, Loyal, At-Risk, Dormant, New)	pages/17_Customer_Intelligence_Engine.py
Product / category / sub-category analysis	pages/20_Product_Sales_Analysis.py
Strategy dashboard, day-of-week heatmaps, festival calendar	pages/30_Sales_Reports_and_Strategy.py
Product catalog viewer	pages/40_Products_catalog.py
MIS viewer (BR_MIS data)	pages/50_MIS_Update.py + services/mis_email_import.py
Lead pipeline + lead entry / status	pages/70_Leads.py + services/imap_lead_import.py
Sales Team Tasks (Daily / Weekly / Monthly / Adhoc)	pages/90_Sales_Team_Tasks.py + services/sales_task_expander.py
Happy Calling list + inline update	pages/95_Happy_Calling.py + services/happy_calling.py
Sales Incentive engine (Tangible 70% + Star 30%)	pages/100_Sales_Manager_Dashboard.py + services/incentive_store.py
Historical (pre FY 26-27) dashboard	pages/old_data_dashboard.py + services/automation.py + services/email_sender.py
Login + role gating	services/auth.py + services/sheets_1.py
Google Sheets read/write (low level)	services/sheets.py
Pending-delivery alert engine (FY 26-27)	services/automation4s.py
WhatsApp deep-link generation	services/automation4s.py (live) / services/automation.py (legacy)
Pending-delivery email (10 AM / 5 PM)	email_job.py + services/email_sender_4s.py
Payment-due email (8 AM / 9 AM)	payment_email_job.py + services/email_sender_4s.py
Happy Calling email (7 AM)	happy_calling_email_job.py + services/email_sender_happy_calling.py
Sales tasks emails (11 AM / 8 PM)	sales_tasks_email_job.py + sales_tasks_status_email_job.py + services/email_sender.py
Monthly performance email (last day, 7 PM)	monthly_performance_email_job.py + services/email_sender_monthly_performance.py
Schedule-Delivery email + Drive invoice attachment	services/email_sender_delivery_schedule.py + services/delivery_readiness.py
Green 'ready for delivery' highlighting	services/delivery_readiness.py
Audit trail of pending-delivery updates	services/delivery_updates.py
Lead email import (every 30 min)	lead_email_import_job.py + services/imap_lead_import.py
MIS daily import (11 AM)	mis_daily_import_job.py + services/mis_email_import.py
Google reviews → 4S sheet star column (22:00)	google_reviews_update_job.py + services/google_reviews_service.py
Right-to-left dashboard ticker	services/dashboard_ticker.py
Number / amount formatting across CRM	utils/helpers.py

CRM feature	Files to open
Cron schedules (timing changes)	.github/workflows/*.yaml
SMTP / Google / GMB credentials	.streamlit/secrets.toml (production), .env (local)

7. Troubleshooting cheat-sheet

Common symptoms and the first file to open.

Symptom you see	First file to look at
A KPI tile on the home page is wrong	pages/b2c_dashboard.py — find the tile, then trace the column to services/sh
A revenue number disagrees between two pages	services/sheets.py (caching / column unification) + utils/helpers.py (formatting
WhatsApp link does nothing	services/automation4s.py (live) — check generate_whatsapp_group_link / ge
Pending Delivery email didn't fire / fired at the wrong time	.github/workflows/godrej-email.yaml (cron in UTC) + email_job.py (SLOT logi
Payment Due email fired twice / wrong time	.github/workflows/payment-email.yaml + payment_email_job.py.
Happy Calling email keeps showing a customer who already got called	services/happy_calling.py — check the keying (ORDER NO fallback to NAME
Sales Team Tasks missing a recurring entry	services/sales_task_expander.py — verify FREQUENCY + week-off map.
MIS green-highlight is wrong on a row	services/delivery_readiness.py + the row in services/mis_email_import.py PO
Customer Intelligence cohort numbers off	pages/17_Customer_Intelligence_Engine.py — cohort thresholds at the top o
Login fails for a user that should work	services/auth.py + services/sheets_1.py (Users sheet).
Incentive screen denies access for an admin	services/incentive_store.py (Incentive_Users sheet, bcrypt).
Lead missing from Leads page	services/imap_lead_import.py (parser) + .github/workflows/lead-email-import.
GMB review stars not appearing in 4S Sales	services/google_reviews_service.py + .github/workflows/google-reviews-fetch
Schedule-Delivery email missing invoice PDF	services/email_sender_delivery_schedule.py — Drive folder path or filename

Appendix — files removed during cleanup (14 May 2026)

Recorded here so you have a written audit trail of what was deleted.

File	Reason
CREDENTIAL_DEBUG_REPORT.md	Old setup doc (.md)
CREDENTIAL_DIAGNOSTIC_CHECKLIST.md	Old setup doc (.md)
FEATURE_SHOWCASE.md	Old release-notes doc (.md)
FIXES_APPLIED.md	Old change-log doc (.md)
GOOGLE_REVIEWS_SETUP.md	Old setup doc (.md)
LEAD_ENHANCEMENTS_SUMMARY.md	Old change-log doc (.md)
QUICK_FIX.md	Old troubleshooting doc (.md)
TROUBLESHOOTING_CHECKLIST.md	Old troubleshooting doc (.md)
streamlit_app/IMAP_SETUP_GUIDE.md	Old setup doc (.md)
streamlit_app/LEADS_PAGE_SETUP.md	Old setup doc (.md)
streamlit_app/MULTI_SOURCE_LEADS_GUIDE.md	Old setup doc (.md)
streamlit_app/TASK_GENERATION_LOGIC.md	Old setup doc (.md)
70_Leads_ENHANCED.py	Orphan duplicate of pages/70_Leads.py
b2c_dashboard_FIXED.py	Orphan duplicate of pages/b2c_dashboard.py
test_calculations.py	Old standalone test, not in any workflow
test_grouping_logic.py	Old standalone test, not in any workflow
streamlit_app/sheets.py	Superseded by services/sheets.py
streamlit_app/automation.py	Superseded by services/automation.py + automation4s.py
streamlit_app/cleanup_unused_sheets.py	One-shot maintenance script
streamlit_app/generate_gmb_refresh_token.py	One-shot OAuth utility (per your call)
secrets.toml.example	Template, not used at runtime (per your call)
streamlit_app/admin_pages/ (5 files)	Not referenced from app.py (per your call)
Google_Places_API_Fix_Guide.docx	BLOCKED — file is currently open in Word; close Word and delete manually.